Project6 - Weather Data Analysis

Step 1:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("DailyDelhiClimateTrain.csv")
df.head()
```

	date	meantemp	humidity	wind_speed	meanpressure
0	2013-01-01	10.000000	84.500000	0.000000	1015.666667
1	2013-01-02	7.400000	92.000000	2.980000	1017.800000
2	2013-01-03	7.166667	87.000000	4.633333	1018.666667
3	2013-01-04	8.666667	71.333333	1.233333	1017.166667
4	2013-01-05	6.000000	86.833333	3.700000	1016.500000

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1462 entries, 0 to 1461
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   date          1462 non-null   str
 1   meantemp      1462 non-null   float64
 2   humidity      1462 non-null   float64
 3   wind_speed    1462 non-null   float64
 4   meanpressure  1462 non-null   float64
dtypes: float64(4), str(1)
memory usage: 57.2 KB
```

```
df.columns
```

Index(['date', 'meantemp', 'humidity', 'wind_speed', 'meanpressure'], dtype='str')

```
df.shape
```

(1462, 5)

```
df.describe()
```

	meantemp	humidity	wind_speed	meanpressure
count	1462.000000	1462.000000	1462.000000	1462.000000
mean	25.495521	60.771702	6.802209	1011.104548
std	7.348103	16.769652	4.561602	180.231668
min	6.000000	13.428571	0.000000	-3.041667
25%	18.857143	50.375000	3.475000	1001.580357
50%	27.714286	62.625000	6.221667	1008.563492
75%	31.305804	72.218750	9.238235	1014.944901
max	38.714286	100.000000	42.220000	7679.333333

Step2: Convert Date Column AND new YEAR and MONTH column

```
df['date'] = pd.to_datetime(df['date'])

df['year'] = df['date'].dt.year
df['month'] = df['date'].dt.month

df.head()
```

	date	meantemp	humidity	wind_speed	meanpressure	year	month
0	2013-01-01	10.000000	84.500000	0.000000	1015.666667	2013	1
1	2013-01-02	7.400000	92.000000	2.980000	1017.800000	2013	1
2	2013-01-03	7.166667	87.000000	4.633333	1018.666667	2013	1
3	2013-01-04	8.666667	71.333333	1.233333	1017.166667	2013	1
4	2013-01-05	6.000000	86.833333	3.700000	1016.500000	2013	1

Step3: Check Missing Values

```
df.isnull().sum()
```

```
date            0
meantemp        0
humidity        0
wind_speed      0
meanpressure    0
year            0
month           0
dtype: int64
```

Step4: NumPy Analysis

```
temp_array = np.array(df['meantemp'])

print("Mean Temp:", np.mean(temp_array))
print("Max Temp:", np.max(temp_array))
print("Min Temp:", np.min(temp_array))
```

Mean Temp: 25.495520655761762
Max Temp: 38.71428571428572
Min Temp: 6.0

🎯 **Why?**

- To understand basic statistics of temperature
- NumPy is faster and optimized for numerical operations
- Shows you can handle **arrays (not just Pandas)**

💬 **Interview Answer:**

"I used NumPy to perform efficient numerical computations like mean, max, and min on temperature data to understand its distribution."

👉 Simple meaning:

- Average temperature kya hai?
- Highest / lowest kya hai?

Step5: Yearly Analysis

```
yearly_temp = df.groupby('year')['meantemp'].mean()
yearly_humidity = df.groupby('year')['humidity'].mean()

print(yearly_temp)
print(yearly_humidity)
```

```
year
2013    24.791494
2014    25.010673
2015    25.114591
2016    27.103373
2017    10.000000
Name: meantemp, dtype: float64
year
2013     63.046292
2014     59.767942
2015     61.430490
2016     58.740174
2017    100.000000
Name: humidity, dtype: float64
```

🎯 **Why?**

- To identify long-term trends
- Helps answer:
  - Temperature increasing over years?
  - Climate change pattern?

💬 **Interview Answer:**

"I performed yearly aggregation to identify long-term trends and observe how temperature and humidity change over time."

👉 Simple meaning: Har saal ka average → trend samajhne ke liye

Step6: Monthly Analysis

```
monthly_temp = df.groupby('month')['meantemp'].mean()
monthly_temp
```

```
month
1     13.311914
2     17.620422
3     22.914103
4     29.376614
5     33.315889
6     33.734011
7     30.995782
8     30.601777
9     30.433216
10    27.113154
11    20.656855
12    15.666317
Name: meantemp, dtype: float64
```

🎯 **Why?**

- To understand **seasonal patterns**
- Helps identify:
  - Summer vs Winter
  - Monsoon impact

💬 **Interview Answer:**

"Monthly analysis helps in identifying seasonal patterns such as temperature variation across different months."

👉 Simple meaning:

- Kaunse month me zyada heat?
- Kaunse month me thand?

Step7: Correlation Matrix

```
corr = df[['meantemp','humidity','wind_speed','meanpressure']].corr()
corr
```

	meantemp	humidity	wind_speed	meanpressure
meantemp	1.000000	-0.571951	0.306468	-0.038818
humidity	-0.571951	1.000000	-0.373972	0.001734
wind_speed	0.306468	-0.373972	1.000000	-0.020670
meanpressure	-0.038818	0.001734	-0.020670	1.000000

🎯 **Why?**

- To find relationships between variables
- Example:
  - Temperature ↑ → Humidity ↓ ?
  - Pressure vs Wind speed?

💬 **Interview Answer:**

"I used correlation analysis to understand relationships between weather variables, which helps in identifying dependencies and patterns."

👉 Simple meaning: Ek cheez badhne se dusri pe kya effect hota hai?

🧩 **SIMPLE SUMMARY (Remember this 🔥):**

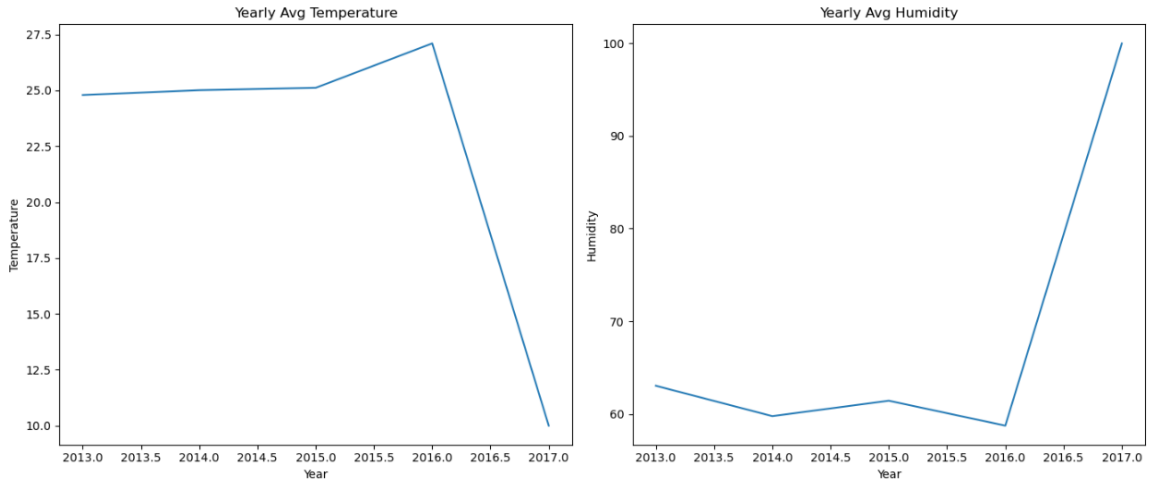
Step	Purpose
NumPy Analysis	Basic stats (mean, max, min)
Yearly Analysis	Long-term trend
Monthly Analysis	Seasonal pattern
Correlation	Relationship between variables

BEST INTERVIEW LINE (USE THIS 🔥)

"I started with basic statistical analysis using NumPy, then performed time-based aggregation (yearly and monthly) to identify trends and seasonality, and finally used correlation analysis to understand relationships between different weather parameters."

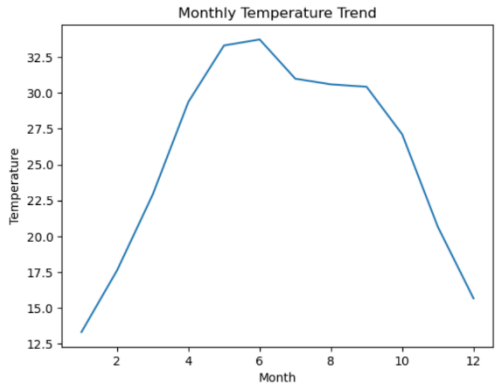
Step8: Multi-Plot (Yearly Trends)

```
plt.figure(figsize=(14,6))

# Temperature
plt.subplot(1,2,1)
plt.plot(yearly_temp.index, yearly_temp.values)
plt.title("Yearly Avg Temperature")
plt.xlabel("Year")
plt.ylabel("Temperature")

# Humidity
plt.subplot(1,2,2)
plt.plot(yearly_humidity.index, yearly_humidity.values)
plt.title("Yearly Avg Humidity")
plt.xlabel("Year")
plt.ylabel("Humidity")

plt.tight_layout()
plt.show()
```



Step9: Monthly Temperature Plot

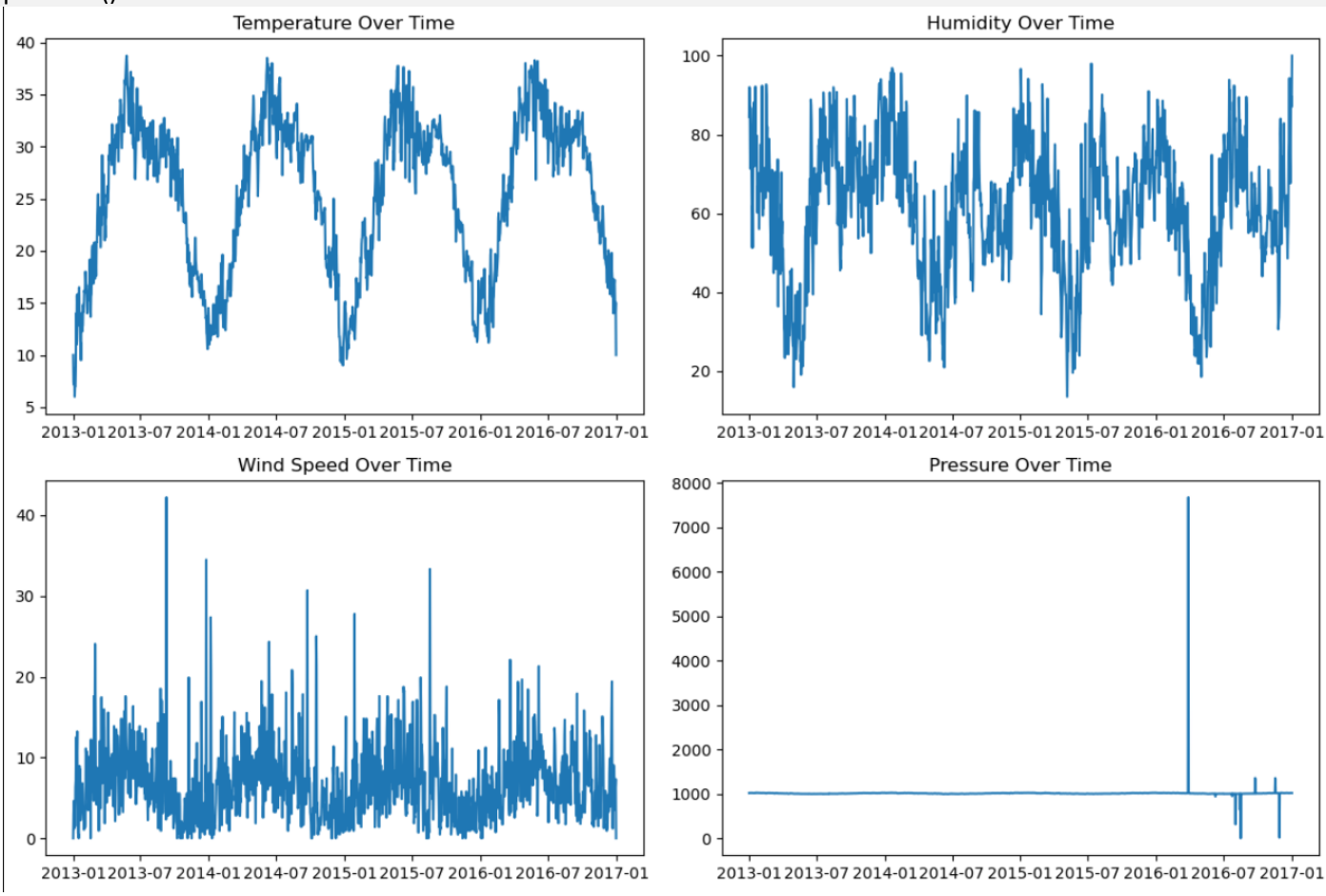
```
plt.plot(monthly_temp.index, monthly_temp.values)
plt.title("Monthly Temperature Trend")
plt.xlabel("Month")
plt.ylabel("Temperature")
plt.show()
```



Step10: Full Multi-Plot (All Features)

```
plt.figure(figsize=(12,8))

plt.subplot(2,2,1)
plt.plot(df['date'], df['meantemp'])
plt.title("Temperature Over Time")

plt.subplot(2,2,2)
plt.plot(df['date'], df['humidity'])
plt.title("Humidity Over Time")
```

```
plt.subplot(2,2,3)
plt.plot(df['date'], df['wind_speed'])
plt.title("Wind Speed Over Time")

plt.subplot(2,2,4)
plt.plot(df['date'], df['meanpressure'])
plt.title("Pressure Over Time")

plt.tight_layout()
plt.show()
```



Step11: Insights (VERY IMPORTANT 💡)

After plots, write insights like:

- Temperature is increasing/decreasing
- Rainfall trend is stable or fluctuating
- Any abnormal year (outlier)

👉 Interviewers LOVE this part.

- Temperature shows seasonal variation (high in summer, low in winter)
- Humidity increases during monsoon months
- Wind speed fluctuates but has no strong trend
- Pressure remains relatively stable

How to Explain in Interview

Say: "I performed weather data analysis using Pandas and NumPy, handled datetime features, and created multi-plot visualizations using Matplotlib to identify yearly temperature and rainfall trends."

OR

"I analyzed time-series weather data by performing date feature engineering, used NumPy for numerical analysis, and created multi-plot visualizations in Matplotlib to identify trends in temperature, humidity, wind speed, and pressure."